Lerche, and Warner had come upon, I was intrigued but also skeptical. The intrigue arose from the realization that if their conjecture was true, it might open a new avenue of research on string theory; the skepticism arose from the realization that guesses are one thing, established properties of a theory are quite another.

During the following months, I thought frequently about their conjecture and, frankly, half convinced myself that it wasn't true. Surprisingly, though, a seemingly unrelated research project I had undertaken in collaboration with Ronen Plesser, then a graduate student at Harvard and now on the faculty of the Weizmarm Institute and Duke University, was soon to change my mind completely. Plesser and I had become interested in developing methods for starting with an initial Calabi-Yau shape and mathematically manipulating it to produce hitherto unknown Calabi-Yau shapes. We were particularly drawn to a technique known as orbifolding, which was pioneered by Dixon, Jeffrey Harvey of the University of Chicago, Vafa, and Witten in the mid-1980s. Roughly speaking, this is a procedure in which different points on an initial Calabi-Yau shape are glued together according to mathematical rules that ensure that a new Calabi-Yau shape is produced. This is schematically illustrated in Figure 10.4. The mathematics underlying the manipulations illustrated in Figure 10.4 is formidable, and for this reason string theorists had thoroughly investigated this procedure only as applied to the simplest of shapes—higher-dimensional versions of the doughnut shapes shown in Figure 9.1. Plesser and I realized, though, that some beautiful new insights of Doron Gepner, then of Princeton University, might give a powerful theoretical framework for applying the orbifolding technique to full-fledged CalabiYau shapes, such as the one in Figure 8.9.

After a few months of intensive pursuit of this idea we came to a surprising realization. If we glued particular groups of points together in just the right way, the Calabi-Yau shape we produced differed from the one we started with in a startling manner: The number of odd-dimensional holes in the new Calabi-Yau shape equaled the number of even-dimensional holes in the original, and vice versa. In particular, this means that the total number of holes—and therefore the number of particle families—in each is the same even though the even-odd